

Port Scanning and Service Enumeration

Lab Objective

The supplied assignment describes an internal, controlled vulnerability-assessment exercise against Metasploitable 2. Its methodology is to confirm reachability, identify open TCP ports, identify services and versions, compare identified software with CVE records, assign CVSS severity where an official score exists, and document evidence, confidence, risk, and remediation. The source report identifies the target as 192.168.65.130 and the environment as VMware NAT / isolated training lab.

Step 1 — Verify the target address

On the Metasploitable VM, confirm its IP address. The supplied report uses 192.168.65.130.

```
ifconfig
```

On Kali, verify that the target responds:

```
ping -c 4 192.168.65.130
```

Kali Linux — Connectivity Check

```
$ ping -c 4 192.168.65.130
PING 192.168.65.130 (192.168.65.130) 56(84) bytes of data.
64 bytes from 192.168.65.130: icmp_seq=1 ttl=64 time=0.003 ms
64 bytes from 192.168.65.130: icmp_seq=2 ttl=64 time=0.003 ms
64 bytes from 192.168.65.130: icmp_seq=3 ttl=64 time=0.003 ms
64 bytes from 192.168.65.130: icmp_seq=4 ttl=64 time=0.003 ms
--- 192.168.65.130 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss
```

Expected result: the host should respond. The supplied report records the target as reachable with low latency.

Step 2 — Initial TCP Port Scan

Start with a basic Nmap scan to identify commonly exposed TCP services.

```
sudo nmap 192.168.65.130
```

Kali Linux — Initial Nmap Scan

```
$ sudo nmap 192.168.65.130
Starting Nmap 7.99
Nmap scan report for 192.168.65.130
Host is up.
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
...
```

Record the open ports. The supplied assessment ultimately identified 23 open TCP ports in its 1,000-port scan.

Evidence from the supplied report

The source scan states that the host was up and that 977 of the standard 1,000 TCP ports were not shown because they were closed/reset. The identified services included FTP, SSH, Telnet, SMTP, DNS, HTTP, SMB, NFS, databases, VNC, Java RMI and other legacy services.

Save your output

```
sudo nmap 192.168.65.130 -oN 01-basic-scan.txt
```

Step 3 — Full TCP Port Scan

Scan the complete TCP range (1–65535) so that services outside the default Nmap port list are not missed.

```
sudo nmap -p- 192.168.65.130
```

Kali Linux — Full TCP Port Scan

```
$ sudo nmap -p- 192.168.65.130
Starting Nmap 7.99
Nmap scan report for 192.168.65.130
Host is up.
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  netbios-ssn
... additional open ports ...
```

```
sudo nmap -p- 192.168.65.130 -oN 02-all-tcp.txt
```

Step 4 — Service and Version Detection

Use Nmap service/version detection to determine what software appears to be listening on each open port. This is important because CVE applicability often depends on the exact product and version.

```
sudo nmap -sS -sV 192.168.65.130
```

Kali Linux — Service & Version Detection

```
$ sudo nmap -sS -sV 192.168.65.130
Starting Nmap 7.99
Nmap scan report for 192.168.65.130
Host is up (0.0033s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1
23/tcp    open  telnet   Linux telnetd
25/tcp    open  smtp     Postfix smtpd
53/tcp    open  domain   ISC BIND 9.4.2
80/tcp    open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
...
8180/tcp  open  http     Apache Tomcat/Coyote JSP engine 1.1
```

The supplied assessment's raw evidence reports examples including vsftpd 2.3.4 on TCP/21, OpenSSH 4.7p1 on TCP/22, Linux telnetd on TCP/23, Apache 2.2.8 on TCP/80, MySQL 5.0.51a-3ubuntu5 on TCP/3306, PostgreSQL 8.3.x on TCP/5432, VNC on TCP/5900, and Apache Tomcat on TCP/8180.

```
sudo nmap -sS -sV 192.168.65.130 -oN 03-service-version.txt
```

Step 5 — OS Detection

Run Nmap's OS fingerprinting against the isolated target.

```
sudo nmap -O 192.168.65.130
```

Kali Linux — OS Detection

```
$ sudo nmap -O 192.168.65.130
Starting Nmap 7.99
Nmap scan report for 192.168.65.130
Host is up.
OS details: Linux / Unix
Service Info: Hosts: metasploitable.localdomain
OSs: Unix, Linux
CPE: cpe:/o:linux:linux_kernel
```

```
sudo nmap -O 192.168.65.130 -oN 04-os-detection.txt
```

The supplied scan evidence identifies the service environment as Unix/Linux and gives a Linux kernel CPE. OS fingerprinting is an inference, so document the result as reported by

Nmap.

Step 6 — UDP Scanning

UDP is connectionless, so UDP scanning behaves differently from TCP scanning and can take longer. Start with Nmap's UDP scan:

```
sudo nmap -sU 192.168.65.130
```

Kali Linux — UDP Scan

```
$ sudo nmap -sU 192.168.65.130
Starting Nmap 7.99
Nmap scan report for 192.168.65.130
Host is up.
PORT      STATE      SERVICE
53/udp    open|filtered domain
111/udp   open|filtered rpcbind
...
UDP scans may take longer because UDP has no TCP-style handshake.
```

```
sudo nmap -sU 192.168.65.130 -oN 05-udp-scan.txt
```

For UDP service/version detection:

```
sudo nmap -sU -sV 192.168.65.130 -oN 06-udp-services.txt
```

Step 7 — NSE / Basic Service Enumeration

Nmap Scripting Engine (NSE) default scripts can collect additional information about discovered services.

```
sudo nmap -sC -sV 192.168.65.130
```

Kali Linux — NSE Enumeration

```
$ sudo nmap -sC -sV 192.168.65.130
Starting Nmap 7.99
Nmap scan report for 192.168.65.130
Host is up.
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1
80/tcp    open  http     Apache httpd 2.2.8
...
Default NSE scripts collect additional service information.
```

```
sudo nmap -sC -sV 192.168.65.130 -oN 07-nse-enumeration.txt
```

Step 8 — Final Combined Scan

After completing the individual steps, perform one consolidated TCP assessment with SYN scanning, version detection, OS detection, and default NSE scripts.

```
sudo nmap -sS -sV -O -sC 192.168.65.130
```

Kali Linux — Final Combined Scan

```
$ sudo nmap -sS -sV -O -sC 192.168.65.130
Starting Nmap 7.99
Nmap scan report for 192.168.65.130
Host is up.
23 TCP services identified in the supplied assessment evidence.
Service/version detection and OS fingerprinting completed.
Save evidence with: -oN FINAL-SCAN.txt
```

```
sudo nmap -sS -sV -O -sC 192.168.65.130 -oN FINAL-SCAN.txt
```

Step 9 — Organize the Evidence

Recommended evidence files:

```
01-basic-scan.t
xt02-all-tcp.txt
03-service-version.t
xt04-os-detection
.txt05-udp-scan.txt
06-udp-services.t
xt07-nse-enumeratio
n.txt08-banners
.txtFINAL-SCAN.
txt
```

Findings Table from the Supplied Assessment

Port	Service	Version / observation	Assessment
23/tcp	Telnet	Linux telnetd; unencrypted administrative traffic	HIGH
80/tcp	HTTP	Apache 2.2.8 (Ubuntu); legacy release	HIGH*
139/445	SMB	Samba 3.x–4.x; exact version needs further probing	HIGH*
1524/tcp	bindshell	Metasploitable root shell exposed without authentication	CRITICAL
3306/tcp	MySQL	5.0.51a-3ubuntu5; exact CVEs depend on build/configuration	HIGH*

